

Om Prasad Nayak

Aspiring Systems Engineer • Learning new things everyday

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🐙 github.com/0-second-shadow-0

🌐 protfolio-swart-tau.vercel.app

SKILLS

- Machine Learning • Deep Learning • Python • Go • Postgres • Rust • Java
- TensorFlow • PyTorch • Terraform • Docker • K8s • OpenTelemetry • Grafana • Redis

PROJECTS

ink-app

📅 Jun 2026 - Present

- Developing a Windows Ink alternative for macOS.
- Building it using Python programming language as it has the best frameworks and libraries for handwriting recognition.
- Utilizing Apple's built-in OCR for a smooth development process, while tackling its limitation regarding non-English language support.
- Integrating EasyOCR finetuned on custom handwriting to fast-track the ML pipeline using Transformers, PyTorch, Accelerate, and Evaluate.

resctl

* Jun 2026 – present

- Developing a cross-platform, real-time system monitoring CLI tool and live terminal dashboard using Rust.
- Engineered low-overhead data ingestion pipelines for multi-resource tracking, including CPU cache, memory, and disk I/O.
- Implementing hardware-level telemetry tracking for GPU VRAM utilization, battery health, and system power draw.
- Optimizing dashboard rendering and data aggregation to ensure minimal CPU and memory footprint during continuous monitoring.

Ultra-bDB

📅 Jun 2026 - Present

- Building a custom B-tree engine featuring ACID transactions and lock-free reads in Go using standard libraries.
- Running concurrency performance benchmarks via `go test -race` to verify lock-free execution safely.
- Profiling CPU and memory efficiency optimizations using the native `runtime/pprof` tooling.
- Enforcing clean, standard practices and high code quality across development using `golangci-lint`.

Email Assistant

📅 Apr 2026 - May 2026

- Implemented multilingual translation capabilities for incoming and outgoing email systems.
- Built clean REST APIs using Spring Boot to enable seamless system integration and backward compatibility.
- Engineered the end-to-end web application using Java, React.js, HTML, CSS, Maven, and Vite.

EDUCATION

B.Tech in Computer Science & Engineering

Silicon Institute Of Technology, Sambalpur

CGPA: 7.14 (till 6th Sem.)

📅 2023 - 2027

Higher Secondary School Certificate (+2 Science)

D.A.V. Public School, PPL Township, Paradeep

Percentage: 72%

📅 2023

Secondary School Examination (Matriculation)

D.A.V. Public School, PPL Township, Paradeep

Percentage: 88%

📅 2021

EXPERIENCES

- Taking part in 5-Day AI Agents: Intensive Vibe Coding Course With Google.
- Participating in Hack2Skill India Run Hackathon 2026 edition.